

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 18 Diversity of Microbial Eukarya

18.1 Multiple Choice Questions

1) Primary endosymbioses gave rise to _____ and secondary endosymbioses gave rise to _____.

- A) chloroplasts in algae / chloroplasts in plants
- B) mitochondria in eukaryotes / chloroplasts in eukaryotes
- C) chloroplasts and mitochondria in plants and red and green algae / chloroplasts in other eukaryotes
- D) chloroplasts and mitochondria in green and red algae / chloroplasts and mitochondria in diatoms

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.1

2) Evidence of _____ supports the hypothesis that eukaryotes are derived from bacteria.

- A) antibiotic resistance
- B) DNA content in the nucleus only
- C) ribosome characteristics only
- D) ribosome characteristics and DNA content in the nucleus

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

3) Which group of *Eukarya* has complicated the phylogenetics of the domain by first being thought of as ancient as compared to the more recent view of them being highly derived eukaryotes?

- A) dinoflagellates
- B) fungi
- C) hydrogenosome-containing eukaryotes
- D) stramenopiles

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.2

4) Which statement best describes our current understanding of the eukaryotic phylogenetic tree?

- A) Ribosomal RNA data allowed the development of the current phylogeny with 6 supergroups.
- B) Whole genome data was used to distinguish the 3 domains.
- C) The most recent analyses have conclusively shown the relationships of the 6 supergroups.
- D) New discoveries will probably lead to changes in the eukaryotic phylogenetic tree.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.2

5) Of the taxa listed, which is NOT classified as a protist?

- A) *Amoeba*
- B) *Euglena*
- C) *Giardia*
- D) *Saccharomyces*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.12

6) Which of the following is NOT a unifying characteristic of diplomonads and parabasalids?

- A) having flagella
- B) lacking chloroplasts
- C) lacking mitochondria
- D) being multicellular

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.3

7) *Giardia* are diplomonads which have nuclei that

- A) appear as two separate chromosomes but are weakly joined.
- B) contain one small and one large nucleus per cell.
- C) have one nucleus enclosed in a mitosome and another nucleus in their cytoplasm.
- D) occur as two similar sizes per cell.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.3

8) Many parabasalids lack _____, which makes them different than most other eukaryotes.

- A) endoplasmic reticula
- B) introns in their genomes
- C) mitochondria and hydrogenosomes
- D) Golgi complexes

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.3

9) A thin, crescent-shaped organism with a single flagellum originating from a basal body and which folds back laterally across the cell would most likely be a member of the genus

- A) *Trypanosoma*.
- B) *Glossina*.
- C) *Entamoeba*.
- D) *Gambiense*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.3

10) A drug that specifically inhibits the replication of the large DNA mass present in large mitochondria would be most effective at killing

- A) euglenids.
- B) ciliates.
- C) *Trypanosoma* spp.
- D) *Toxoplasma* spp.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.3

11) Which group of eukaryotes LACKS alveoli?

- A) apicomplexans
- B) ciliates
- C) dinoflagellates
- D) oomycetes

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.5

12) Which statement BEST describes the habitats of the fungi?

- A) Most fungi are aquatic, microscopic, and primarily freshwater.
- B) Most fungi are aquatic and primarily marine.
- C) Most fungi are terrestrial, microscopic, and inhabit soil or dead plant matter.
- D) Most fungi are terrestrial and macroscopic.

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 18.8

13) Oomycetes ("water molds") contain _____ in their cell walls, while most fungi have _____ in their cell walls.

- A) cellulose / chitin
- B) chitin / cellulose
- C) silica / chitin
- D) cellulose / silica

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.6

14) Which statement is TRUE regarding fungal spores?

- A) Conidia and ascospores are sexual spores.
- B) Ascospores and basidiospores are asexual spores.
- C) Basidiospores and conidia are sexual spores.
- D) Ascospores and basidiospores are sexual spores.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.9

- 15) Which statement is TRUE regarding the ecological role of plant degradation by fungi?
- A) In brown rot the cellulose is attacked preferentially, and the lignin is left unmetabolized.
 - B) In brown rot the lignin is attacked preferentially, and the cellulose is left unmetabolized.
 - C) In white rot the cellulose is attacked preferentially, and the lignin is left unmetabolized.
 - D) In white rot the lignin is attacked preferentially, and the cellulose is left unmetabolized.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.8

- 16) Diatoms are
- A) found only in freshwater.
 - B) more closely related to slime molds than to algae.
 - C) usually morphologically symmetric.
 - D) multicellular phototrophs.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.5

- 17) Which algae are most closely related to plants?

- A) red algae
- B) green algae
- C) brown algae
- D) euglenoids

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

- 18) Endolithic phototrophic communities are most commonly found in _____ environments.

- A) dry
- B) hot
- C) humid
- D) windy

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

- 19) What would a fucoxanthin carotenoid stain be most useful in identifying?

- A) brown or golden algae
- B) flagellated algae
- C) photosynthetic diatoms
- D) slime molds from other molds

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.5

20) A geologist going through a sediment core is likely to find remnants of ancient calcium carbonate containing _____ and silica containing _____.

- A) diatoms / chlorarachniophytes
- B) foraminiferans / diatoms
- C) chlorarachniophytes / foraminiferans
- D) foraminiferans / chlorarachniophytes

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.6

21) What phenomenon accounts for the appearance of chloroplasts in some unicellular phototrophic eukaryotes?

- A) transformation of hydrogenosomes
- B) secondary endosymbioses of green or red algae
- C) primary endosymbioses of green or red algae
- D) horizontal gene transfer

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

22) Which group of amoebozoans is most likely to cause illness in humans?

- A) dictyostelium
- B) entamoebas
- C) gymnamoebas
- D) slime molds

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.7

23) How can the amoebozoans and radiolarians be distinguished from each other with microscopy?

- A) Amoebozoans have lobe-shaped pseudopodia, and radiolarians have threadlike pseudopodia, and radiolarians have radial symmetry.
- B) Amoebozoans have lobe-shaped pseudopodia, amoebozoans have radial symmetry, and radiolarians have unusually large, thick pseudopodia.
- C) Amoebozoans have large but thin pseudopodia and radiolarians have thick, lobe-shaped pseudopodia.
- D) Amoebozoans have radial symmetry and threadlike pseudopodia, whereas radiolarians have lobe-shaped pseudopodia.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.6, 18.7

24) Which microbial group is responsible for massive amphibian die-offs worldwide?

- A) cercozoans
- B) ciliates
- C) chytrids
- D) zygomycetes

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.10

25) Which of the following is NOT a member of the *Eukarya*?

- A) protists
- B) fungi
- C) spirochetes
- D) red algae

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

26) The function of the parabasal body is to

- A) allow for photosynthesis to occur.
- B) give structural support to the Golgi complex.
- C) allow for the formation of energy from aerobic respiration.
- D) give structural support to the endoflagellum.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.3

27) The amoebzoa use what structures for movement and feeding?

- A) threadlike pseudopodia
- B) cilia
- C) flagella
- D) lobe-shaped pseudopodia

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.7

28) The amoebzoa that are parasitic for humans are the

- A) gymnamoebas.
- B) radiolarians.
- C) entamoebas.
- D) oomycetes.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.7

29) One of the earliest diverging lineages of fungi is the

- A) chytridiomycetes.
- B) zygomycetes.
- C) glomeromycetes.
- D) ascomycetes.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.9

30) A fungal mycelium is a large accumulation of _____, which may appear dusty or fuzzy due to the presence of _____.

- A) coenocytes / conidia
- B) coenocytes / mycorrhizae
- C) hyphae / conidia
- D) hyphae / mycorrhizae

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

31) What are the most abundant biomolecules in a typical fungal cell wall?

- A) amino acids
- B) carbohydrates
- C) lipids
- D) porins and channels

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

32) Which spores are NOT involved in sexual reproduction of fungi?

- A) ascospores
- B) basidiospores
- C) conidia
- D) zygosporangia

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

33) In contrast to bacterial endospores, fungal spores have substantially less resistance to

- A) chemicals.
- B) desiccation.
- C) heat.
- D) freezing.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.8

34) The unusual ability of _____ being flagellated and motile enables _____.

- A) ascospores / their dispersal in aquatic systems
- B) ascospores / spreading of those pathogens in blood
- C) zoospores / dispersal in aquatic systems
- D) zoospores / spreading of those pathogens in blood

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 18.10

35) Spoilage of bread, which appears dark green-blue to black in color, is most likely caused by the fungus

- A) *Aspergillus niger*.
- B) *Batrachomyces dendrobatidis*.
- C) *Glomus* spp.
- D) *Rhizopus nigricans*.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.11

36) Obligate symbiotic fungi that form associations with plants by entering their cortical cells without invading the plant cell's membrane are called _____ and include the genus _____.

- A) ectomycorrhizae / *Agaricus*
- B) ectomycorrhizae / *Glomus*
- C) endomycorrhizae / *Agaricus*
- D) endomycorrhizae / *Glomus*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.11

37) Microsporidia were previously thought to be an early-branching taxon of *Eukarya* due to their parasitic lifestyle. However, new evidence suggests that they are more closely related to the

- A) ascomycetes.
- B) chytridiomycetes.
- C) glomeromycetes.
- D) zygomycetes.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.10

38) Ascospores are produced in all of the following taxa EXCEPT

- A) *Aspergillus*.
- B) *Candida*.
- C) *Rhizopus*.
- D) *Saccharomyces*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.12

39) For a pure culture of *Saccharomyces cerevisiae* cells that were all derived from a single cell to mate

- A) a mating-type switch in one or more cells in the culture must occur.
- B) both the α and the a gene must be fully expressed in the initial population.
- C) the α gene must be fully repressed in the population.
- D) some cells must exist as diploids.

Answer: A

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 18.11

40) The majority of basidiomycetes living in soil occur as

- A) diploid mycelia.
- B) diploid basidiocarps.
- C) haploid mycelia.
- D) haploid basidiocarps.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.13

41) The accessory pigment phycoerythrin is dominant in the _____ group of algae, which mostly lack flagella, have multicellular lifestyles, and occur in marine waters. Those in particular which deposit calcium carbonate in coral reef systems are classified as _____ algae.

- A) chlorophyte / coralline
- B) chlorophyte / rhodolith
- C) rhodophyte / coralline
- D) rhodophyte / rhodolith

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.14

42) What are the two main groups of green algae?

- A) chlorophytes and chrysophytes
- B) chlorophytes and charophytes
- C) chrysophytes and charophytes
- D) chlorarachniophytes and chlorophytes

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

43) In a freshwater pond, a green alga is likely to be

- A) only colonial.
- B) only filamentous.
- C) only unicellular.
- D) colonial, filamentous, multicellular, or unicellular.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

44) The green alga *Botryococcus braunii* is a candidate for production of biofuel due to its ability to produce

- A) and withstand large amounts of alcohols.
- B) hydrogen gas.
- C) long-chain (C30-C36) hydrocarbons.
- D) short-chain (C8-C14) hydrocarbons.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

18.2 True/False Questions

1) The DNA in chloroplasts and mitochondria of eukaryotic cells is derived from *Bacteria*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

2) Divergence of the green and red algal lineages at least partially explains how two different primary endosymbiosis events occurred to form different chloroplast types in different taxa.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

3) Mitochondria and chloroplasts have their own DNA but use the ribosomes of the host cell.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

4) Although ribosomal RNA has been useful in identifying the three domains, ribosomal RNA gene sequences have not been sufficient to develop accurate eukaryotic phylogenies.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.2

5) The endosymbiosis that formed mitochondria likely occurred after eukaryotes developed photosynthetic capabilities.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.2

6) All ciliates have cilia at some point during their life cycle, and some use the cilia not just for movement but also to direct food as well.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.4

7) Among protists, only the ciliates have a micronucleus and macronucleus where genes are located in two discrete areas within each cell for redundancy.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.4

8) Dinoflagellates have flagella that enable motility and occur in both freshwater and marine waters.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.4

9) All mushrooms are basidiomycetes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.13

10) Sporozoites observed in an insect's body likely indicate infection by an apicomplexan.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.4

11) Despite oomycetes being called "water molds," they are phylogenetically distant from most other fungi.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.5

12) All Stramenopiles are phototrophic microorganisms.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.5

13) Some chrysophytes (or "golden algae") are actually chemoorganotrophs rather than phototrophs.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.5

14) Silica is a characteristic component of diatom and oomycete cell walls.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.5

15) Foraminifera that form test structures are exclusively found in marine waters.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.6

16) Slime molds feed by phagocytosis and are more closely related to amoebas than to fungi.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.7

17) Hyphae are tubular cell walls that surround the cytoplasmic membrane.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

18) There are two different mating types in the yeast *Saccharomyces cerevisiae*, and they can switch from one type to another.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.12

19) Yeasts are immature fungal spores.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

20) Most fungi are aerobic chemoorganotrophs that decompose large molecular weight compounds by secreting extracellular enzymes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

21) A polymer of N-acetyl-glucosamine, chitin, is a primary component to most fungal cell walls.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

22) The chlorophytes use both chlorophylls *a* and *b* for photosynthesis, which make them appear green.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

18.3 Essay Questions

1) How does the genus *Pfiesteria* cause problems in both humans and fish?

Answer: The toxic dinoflagellate *P. piscicida* produces spores containing neurotoxins that can kill the skin of fish. The lesion is then subject to opportunistic pathogens that cause more diseases, often resulting in death. The neurotoxins also affect humans by giving rashes to swimmers as well as respiratory problems.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.4

2) Discuss three findings that support the theory of primary endosymbiosis.

Answer: Possible answer topics could include any three of the following: antibiotics that disrupt the 70S ribosomes of *Bacteria* also disrupt protein synthesis in chloroplasts and mitochondria; chloroplast and mitochondrial ribosomes are 70S just as other prokaryotes; eukaryotic nuclei contain genes derived from *Bacteria*; hydrogenosomes are thought to be divergent mitochondria and they, too, contain genes found in *Bacteria*; mitochondria and chloroplasts contain DNA in the same form as bacteria (covalently closed circular DNA); and rRNA gene sequencing group chloroplasts and mitochondria with *Bacteria*.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

3) What is an endolithic algal community? Where are these found, and how do they survive?

Answer: Endolithic algal communities would likely be found on rocks in aquatic environments. The rock communities can contain a coexistence of algae, fungi, and lichens. As the rocks become weathered, more pores are formed, and more surface area is exposed for photosynthetic microorganisms to thrive using light (photons) to obtain energy.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

4) Describe the differences between the algae and the cyanobacteria.

Answer: Algae are photoautotrophic eukaryotes, whereas cyanobacteria are bacteria. The principle difference is algae use nuclei to store their genetic material, while cyanobacteria lack nuclei.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.15

5) Compare and contrast the life cycles of the cellular and the acellular slime molds.

Answer: Acellular (or plasmodial) slime molds have plasmodial vegetative forms that contain diploid nuclei. Cellular slime molds have haploid nuclei in their vegetative form. Both are motile in their vegetative forms and obtain nutrition through engulfing particles. When conditions are favorable, both also produce fruiting bodies that differentiate into spores.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.7

6) Discuss the theory of secondary endosymbiosis and defend the theory by providing evidence that supports it.

Answer: Secondary endosymbiosis occurred with either a green or red algal cell (which was the result of primary endosymbiosis) that was engulfed by a nonphototrophic eukaryote. This process explains how chlorarachniophytes and euglenids obtained chloroplasts and became photosynthetic. Secondary endosymbiosis accounts for the great phylogenetic diversity among phototrophic eukaryotes. There is also morphological and sequence-based evidence that support this theory.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 18.1

7) Describe the mutualistic relationship that arbuscular mycorrhizae form with plants.

Answer: Approximately 80% or more of all land plants form a mutualistic relationship with arbuscular mycorrhizae. The hyphae of these fungi form swollen vesicle invaginations (or arbuscules) inside plants' roots. The surface area of the roots increases, which allows for the plants to obtain more nutrients in the soil. The arbuscular mycorrhizae (fungi) benefit from fixed carbon provided by the plant.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.8

8) Compare and contrast the ectomycorrhizae and the endomycorrhizae.

Answer: Ectomycorrhizae do not deeply embed themselves into plant roots; instead, they appear to form sheaths around the roots. Endomycorrhizae do penetrate extensively into plant roots.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 18.8

9) Chytridiomycosis is thought to be associated with increases in environmental temperature associated with global warming. Design a series of experiments that would help to determine whether this hypothesis is likely to be correct.

Answer: Answers will vary, but the overall experimental design should involve replicate trials where temperature is the sole variable. The chytridiomycosis disease in frogs would need to be quantified in some way so that comparisons could be drawn with respect to temperature changes.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 18.10

10) Consider the development of a fossil record for a watershed. A current inventory of all species already has been taken. Which microorganisms might you first consider looking for with the objective of developing a microbial fossil record? Why? Also, what might depth profiling accomplish?

Answer: Answers will vary, but diatoms are excellent for developing a fossil record of a watershed. Their glasslike cell structure (composed of silica) should remain intact for a long time, and dead diatoms will accumulate in the sediments over time. Assuming that no major changes to the sediment have occurred over time, looking at diatom diversity at different depths could show how the diatom community has changed over time.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 18.5